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CAPNOGRAPH MAINTENANCE AND CALIBRATION: WHAT YOU NEED TO DO

by Krunal Prajapati



Capnograph maintenance is simpler than most people fear, and more important than most people realise. A capnograph is a measuring instrument. If it drifts, it does not stop working. It keeps giving you numbers, and they are quietly wrong.

The good news is that modern devices need very little. The catch is that the small amount they do need is often misunderstood, starting with the difference between zeroing, calibration and a span check. This guide sorts that out and gives you a practical schedule.

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Key takeaways

- Zeroing, calibration and a gas span check are three different things.
- Most modern NDIR capnographs need **no routine calibration**, but they do need correct zeroing.
- Never zero the device while it is connected to a patient, and keep your own breath away from it.
- A periodic gas span check verifies accuracy against a known CO2 concentration.
- Sidestream devices need more upkeep, because of the sampling line and water trap.

Zeroing, calibration and span check are not the same thing

This is the confusion that causes most of the trouble.

Procedure	What it does	How often
Zeroing	Tells the device what zero CO2 looks like, using room air, and adapts it to the optical characteristics of the adapter fitted	Before use, when changing adapter type, and whenever readings look offset

Procedure	What it does	How often
Calibration	Adjusts the device's measurement scale against reference gases	Most modern NDIR devices do not need this routinely. Follow the manufacturer
Gas span check	Verifies the reading against a known CO2 concentration, to confirm it is still accurate	Periodically, per the manufacturer's schedule

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The distinction matters. Zeroing is a routine, daily thing you do. A span check is a periodic verification. Full calibration is usually a service task, and on many modern devices, not needed at all.

How to zero a capnograph correctly

Zeroing sounds trivial and is easy to get wrong.

The device needs to see genuine room air, which contains almost no carbon dioxide, roughly 0.04%. A patient's breath contains around 4 to 5%, more than a hundred times as much. If the device zeroes on anything but clean air, every reading afterwards is shifted.

- **Never zero while connected to the patient.** Remove the adapter or cannula from the patient first.

- **Keep all CO2 sources away**, including your own breath. Carbon dioxide is heavier than air and will settle around the sensor if you lean over it.
- **Zero with the adapter fitted**, not bare. On a mainstream device, the zero adjusts the sensor to the optical properties of that adapter, so it must be in place.
- **Re-zero when you change adapter type**, and whenever the readings look offset.
- **Let it finish.** Do not connect until initialisation and autozero are complete.

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For where the sensor should sit, see capnography sensor placement.

Does a capnograph need routine calibration?

Often, no. This surprises people.

Many modern NDIR capnographs are solid-state and factory-calibrated, and the manufacturer states that no routine calibration is required. The reference channel inside the sensor corrects for drift as the light source ages, which is a large part of why these devices stay accurate for years. See how NDIR capnography sensors work.

What replaces routine calibration is **verification**. You periodically confirm the device is still reading accurately, and only if it fails do you re-zero and, if it still fails, call service. Always follow the schedule in your own device's manual rather than a general rule.

The gas span check

This is the verification step.

A calibration gas cylinder containing a known concentration of carbon dioxide is flowed through the airway adapter, and you confirm the device reads that concentration within its stated accuracy.

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The sequence is straightforward. Confirm the device reads zero first, re-zeroing if needed. Fit a new airway adapter. Flow the calibration gas through it. Check the reading falls within the accuracy specification. If it does not, re-zero and repeat. If it still fails, the device needs service.

How often depends on the manufacturer's recommendation, and on how heavily the device is used.

Cleaning and adapters

Handle the optical path carefully, because that is where accuracy lives.

Disposable airway adapters are single-patient use. Discard them as biohazardous waste after use. They remove any sterilisation burden and any risk of cross-contamination, which is why they are common.

Reusable adapters, where used, must be cleaned strictly according to the manufacturer's instructions, and the windows must be left completely clear. A film on the window changes what the light sees.

The sensor itself is wiped as the manufacturer directs. Never immerse it, and never scratch or scrub the optical windows.

Sidestream needs more upkeep

f you run a sidestream or microstream device, add hese.

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Water trap. It fills, and it must be emptied and correctly eated. A loose trap creates a vacuum leak, which shows up as a flat line or an artificially low reading.

Sampling lines. Single use. Replace them when they occlude, and do not try to clear a blocked line.

Pump and filters. These are service items on some devices. Check the manual.

A mainstream device has none of this hardware, so this whole section does not apply to it. See mainstream vs sidestream capnography.

Batteries and physical care

- Keep the battery charged, and replace it per the manufacturer's guidance. A portable device is no use flat.
- Protect the sensor and cable. Cables fail from being pulled, trodden on and rolled over.
- Store the device somewhere clean and dry, not loose in a drawer with metal instruments.
- Keep it away from high-frequency electrosurgical equipment and MRI, which most manuals warn against.

A practical maintenance schedule

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When	What to do
Before each patient	Zero in clean room air with the adapter fitted. Check the windows are clear. Fit a fresh adapter
After each patient	Dispose of the single-use adapter. Wipe the sensor per the manual
Daily	Check the battery, cable and sensor for damage. Sidestream: empty the water trap
Periodically, per the manual	Gas span check to verify accuracy
Annually, or per the manual	Manufacturer service check
Whenever readings look wrong	Re-zero first. If it persists, span check. If it still fails, call service

When to call service

Do not keep using a device you do not trust.

Call service if the readings remain offset after re-zeroing, if it fails a gas span check, if the sensor or cable is physically damaged, or if the device throws errors it should not. A monitoring device that is wrong is worse than one that is obviously broken, because you will act on what it tells you. See capnography troubleshooting to rule out the simple causes first.

Where RespiCOz fits

Maintenance is one of the quiet reasons the choice of device matters.

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RespiCOz is a **mainstream** capnograph, so the entire sidestream upkeep list does not apply. There is no sampling line to replace, no water trap to empty and re-seat, and no aspiration pump to service. Day-to-day maintenance is short: zero it properly, keep the adapter windows clear, look after the cable and battery.

It is also serviced locally. RespiCOz is **made in India** and backed by a **dedicated technical team**, not a distributor, so a span check that fails or a sensor that needs attention does not mean waiting weeks for an imported part. It carries a **two-year device warranty** and a one-year sensor warranty, and it is CDSCO-approved, so the manufacturer sits inside India's post-market surveillance framework.

For why that matters more than most buyers realise, see warranty, service and support for medical devices in India. Always follow the maintenance schedule in your own device manual.

Ready to buy? Request a quote for your hospital here.

Frequently asked questions

How often should a capnograph be calibrated? Many modern NDIR capnographs require no routine calibration at all, because they are solid-state and factory-calibrated with an internal reference channel. What they need instead is correct zeroing before use and a periodic gas span check to verify accuracy. Follow your device's manual.

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What is the difference between zeroing and calibration? Zeroing tells the device what zero CO2 looks like, using room air, and takes seconds. Calibration adjusts the measurement scale against reference gases and is usually a service task. A span check simply verifies the device is still accurate.

Can I zero a capnograph while it is attached to the patient? No. Remove the adapter or cannula from the patient first, and keep your own breath away from the sensor. Zeroing on exhaled gas sets the baseline wrong and shifts every reading afterwards.

How do I clean a capnography airway adapter? Most airway adapters are single-patient use and are disposed of as biohazardous waste. Reusable adapters must be cleaned exactly as the manufacturer directs, with the optical windows left completely clear.

What maintenance does a mainstream capnograph need? Very little. Zero it correctly, keep the adapter windows clean and clear, care for the cable and battery, and run a gas span check periodically. There is no sampling line or water trap to maintain.

Conclusion

Capnograph maintenance comes down to a short list. Zero the device properly in clean air, with the adapter fitted and away from the patient. Keep the optical windows clear. Verify accuracy periodically with a gas span check. Look after the cable, battery and sensor. And call service when a reading stays wrong after re-zeroing.

Most modern devices need no routine calibration, which makes the little maintenance they do need all the more

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Perform regular calibration. A capnograph that drifts does not warn you. It just tells you the wrong number confidently. For setup, see capnography sensor placement. For faults, see capnography problems.

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